1. The verification contract

The program is a verifier of exact mathematical statements, not an oracle for Hilbert's Tenth Problem over $\mathbb{Q}$. A positive row is accepted only after its stated hypotheses and identities have been replayed. A bounded scan licenses a statement about its rows and no statement about an unscanned complement. By contrast, the L22 fixed-Z elimination, the L23 half-sieve theorem, the L24 dyadic diagonal obstruction, L25's parity split, L26's reciprocal/local identities, and L27's dyadic/target character theorem are symbolic proofs or finite exhaustions with their uniform arguments displayed separately. Every scan or search remains separately labelled. The L23 lower bound concerns small-prime-clean progression members only, not fully emergent-free members. Classical Schinzel H remains the sole conjectural input to the established member-existence chain.

1.1. **The label calculus.** The ledger uses four labels as a small calculus of admissible inference.

PROVED means that an exact implication, identity, certificate, or finite exhaustion was replayed. The L22 endpoint–Newton elimination, the L23 dimension-1/2 beta-sieve lower bound and parity barrier, the L24 emptiness of both unscaled diagonal orientations on $\Phi(\mathbb{Q}_2)$, L25's complementary parity escape, L26's reciprocal identity, and L27's one-piece dyadic and aligned-target local theorem are PROVED at their stated scopes. A finite closure such as $[179, [-1, 1]]$ remains PROVED only for that row.

CONDITIONAL means that every remaining hypothesis is stated and named. All canonical-branch Schinzel admissibility conditions, including fixed-fiber irreducibility and admissible-a selection, are PROVED. Classical Schinzel H alone is then assumed to produce simultaneous prime values and hence ladder-zero members. The six-unknown conclusion is conditional on Schinzel H and on no additional per-cell certificate.

EVIDENCE means a measured statistic with its sampling frame attached. The matched obstruction artifact reports marginal ratio 0.9993, a union residual of $-2.9/8760$, and a marginal-preserving permutation statistic $z = -0.09$ with empirical tails 0.61/0.53. Likewise, the L24 search's 4,026,282 zero hits corroborate the diagonal theorem but cannot prove emptiness. These numbers validate their recorded finite frames as EVIDENCE; they license neither an unscanned density theorem nor a uniform no-point claim.

OPEN names an unproved statement and is never filled by a scan. The unconditional simultaneous prime-value assertion and the fixed-family two-large-divisor detector asymptotic remain OPEN. L28 closes the L26 quartic trace field and bad-sign squareclass; L29 closes the distinct reciprocal-lift squareclass; and L31 proves one globally good reciprocal member at $w = 13$, not a uniform member theorem. L32 removes base selection at 2 and w, but not emergent numerator/denominator primes. Cube-compatible controlled rational points on L30's lower-degree even quartic remain OPEN. L25's own aligned-target and $w = 3, 5$ questions remain open for those two scaled formulas, although L27 and L30 solve them on different sections. The unscaled diagonal five-count is not in this list: it is PROVED FALSE because both unit orientations are Φ-empty over $\mathbb{Q}_2$.

Labels only move upward through a valid proof. In particular, `PrimalityBound`, `FactorBudget`, a timeout guard, a zero-Jacobi outcome, or a missing producer cannot upgrade EVIDENCE to PROVED and cannot turn an open case into a negative result. A refusal contributes *unknown*; only a Fermat witness, a proper factor, or another exact certificate contributes a mathematical verdict.

2. Proven-primality arithmetic

2.1. **The deterministic range.** The entry point `_is_prime` first applies strong Miller–Rabin at the thirteen bases from 2 through 41 [3]. Its strict deterministic cutoff is

$$\texttt{\_MR\_LIMIT} = 3317044064679887385961981.$$

For n below this bound, passing all bases is a proved primality verdict. At or above the bound, passing is only a screen and the code proceeds to a certificate; it never accepts "probable prime" as prime.

This distinction is load-bearing because factorization feeds exact Hilbert symbols and ramification sets. The regression suite therefore includes the least strong pseudoprime at the cutoff, a prime above the cutoff that must be Pocklington-certified, hard cases that must refuse, and a balanced semiprime that must exhaust the factor budget rather than be guessed.

2.2. **Pocklington and the relaxed threshold.** For a screened n, the engine writes

$$n - 1 = FR,$$

where every prime dividing the accumulated part F is recursively certified. For each such prime q, it seeks a base a satisfying

$$a^{n-1} \equiv 1 \pmod{n}, \qquad \gcd\bigl(a^{(n-1)/q} - 1, n\bigr) = 1.$$

The generalized Pocklington conclusion is immediate when $F^2 > n$. A failed Fermat congruence or a proper gcd proves compositeness; failure to find a usable witness is a `PrimalityBound` refusal.

Before attempting harder splitting, the implementation trial-divides $n-1$ by every prime below 10^6. It computes the integer ceiling of $n^{1/3}$ exactly and stops that trial loop as soon as the accumulated certified part reaches the cube-root threshold. Uncertifiable recursive chunks and factor-budget-exhausted chunks are excluded from F, never counted as if they had factored.

At the weaker threshold $F^3 \geq n$, the Brillhart–Lehmer–Selfridge relaxation, in the form recorded as Crandall–Pomerance Theorem 4.1.5, applies. The code first checks the required coprimality of F and R, writes $R = c_2F + c_1$, and tests

$$D = c_1^2 - 4c_2.$$

Under the certified congruence conditions a composite n can then have exactly two factors of the form $uF + 1$ and $vF + 1$. A nonnegative square D with the correct parity reconstructs those two factors and proves compositeness; otherwise the discriminant test proves primality. This is a certificate theorem, not a heuristic strengthening of Miller–Rabin.

2.3. **Explicit refusal types.** `PrimalityBound` means that the screen passed outside its deterministic range but the certified part or witnesses were insufficient. `FactorBudget` means Brent's rho routine exhausted its bounded work on a large remaining composite. Callers either propagate these exceptions or mark the candidate as refused. The suite contains adversarial paths specifically to ensure that a partial factorization is not returned as complete and that a refusal is not counted as a soluble member.

3. Suite architecture and frozen authority

A default invocation is `python3 h10q.py`; the extended invocation adds `--extended`. Both begin with arithmetic self-tests and the ledger verification, then replay the successive finite layers. The current engine note records approximately 51 seconds for the default suite and approximately 118 seconds for the extended suite on the recorded run.

The difference is scope, not soundness. The default L6 replay covers the odd prime targets below 100, while the extended replay reaches the full 61-row authority below 300. L13 class sampling uses 12 escape rows by default and 40 in the extended run; fully soluble escape witnesses use 24 versus all 60. Most importantly, the default L14 check takes a fixed seeded sample of 60 frozen H-class members, while the extended check replays all 293.

The first two authorities have a stronger serialization contract. Each suite run derives canonical JSONL text from the in-code objects and requires `data/l6_witnesses.jsonl` and `data/l9_`

Table 1. Frozen finite authorities and their intended role.

Authority	Rows	Role
`_L6_WITNESSES`	61	bounded canonical tied evidence
`_L9_STEERED`	345	steered cell witnesses, fully replayed
`_L11_CLASSES`	190	reachable aligned prime classes
`_L12_ESCAPE`	103	non-coprime aligned escape classes
`_L13_ESCAPE2`	60	fully soluble walled-cell witnesses
`_L13_RESIDUAL`	6	legacy pointwise audit slice
`_L13_ZERO_BAD`	16	frozen ladder regression rows
`_L13_H_CLASSES`	293	one certified member per aligned grid class

`steered.jsonl` to match byte for byte. Drift, absence, or an underived field fails the run. The only supported export command is

```
python3 h10q.py --export-evidence
```

which writes those canonical serializations; it does not canonize arbitrary search logs.

4. Artifact schemas and licenses

A schema is part of the proof boundary. Row types say whether a record is configuration, a replayable certificate, a failed attempt, or an aggregate. The principal persisted objects are summarized below; "licenses" always means "licenses at the recorded finite scope."

4.1. **Reading rows without overclaiming.** Metadata is a scope declaration, not a certificate by itself. A `meta` row identifies the authority, coordinate convention, engine, window, and generator assumptions. The claim-bearing payload is then found in a class, canonical, kernel-replay, or closure record, while a summary is a deterministic reduction of those payload rows. Consequently a reader should not cite a global total without retaining the class-level artifact from which it was reduced, and should not cite an attempted class as a certified class.

The canonical closure document makes this lineage especially explicit. Its record stores the cell and family, the aligned progression $Q = q_1 + k_{\text{zero}}N$, the resulting member parameters, the ladder rung, and the source scan. The Python authority stores the corresponding tuple needed for replay. During L14 verification the suite reconstructs Q, proves it prime, forms $b = \varepsilon f Q$, re-runs the smooth emergent test and cofactor ladder, and accepts only the verdict `zero`. The optional full tied and ramification calculations are cross-checks rather than hidden prerequisites for this accepted certificate.

Large integers and rational parameters are serialized as decimal strings or explicit numerator–denominator data where the producing schema requires them. Status words such as `bad`, `deferred`, `nonprime`, and `refused:FactorBudget` are intentionally not coerced to booleans. This preserves the distinction between a proved negative result, a skipped member, and an arithmetic refusal. Schema compatibility therefore includes both field names and status semantics; a consumer that collapses those states has changed the estimand even if its JSON parser succeeds.

The horizon cross-k file is deliberately different. Its meta row freezes the probe space and determinism contract; untagged rows record primality, tied, ramified, and size outcomes. For H109 it contains 1,275 probes through $k \leq 50$, but no row with both requested auxiliary successes. That is an auxiliary finite non-hit, not a defect in the accepted zero-bad closure.

5. Exact reproduction protocol

Run from the repository directory `math/h10q`. The downstream matched chain is:

```
python3 l17_ratemodel_padic.py
python3 l17_badroots_closures.py
python3 l17_matched.py
```

The middle command imports the checked-in p-adic module and rebuilds closure- authority roots from the persisted sieve; the final command consumes those roots and the same persisted member rows. Thus the first arrow is also a code dependency. It does not claim recovery of the original session-side producer for the older censored-rate artifact.

The independent class-risk and Schinzel audits are regenerated by

```
python3 l17_classrisk.py
python3 l17_schinzel_audit.py
```

The former writes `data/l17_classfactors.jsonl` and `/tmp/l17_classrisk.report`; the latter writes `data/l17_schinzel_audit.jsonl` and its report under `/tmp`.

The established off-grid horizon is replayed one cell at a time:

```
python3 l17_horizon101.py
python3 l17_horizon103.py
python3 l17_horizon107.py
python3 l17_horizon109.py
python3 l17_horizon109_crossk.py
```

Each closure generator overwrites its named data artifact and emits or writes a report; the cross-k generator emits no elapsed or host-dependent fields. The ledger records about 17 minutes for that 1,275-probe auxiliary scan and records byte-identical regeneration for it. No runtime is supplied here for commands for which the ledger records none.

The L19/L20 chain is regenerated from `math/h10q` by

```
python3 l19_tauzero.py
python3 l19_classexist.py
python3 l19_cell179.py
python3 l20_admissible.py
```

Each command writes its correspondingly named JSONL artifact under `data/` and a human-readable report under `/tmp`. The first two replay the wall and uniform class constructions, the third replays the finite member closure, and the fourth records the L20 admissibility boundary subsequently sharpened by L21.

Run the L21 chain from the repository root with the exact commands

```
nice -n 19 python3 math/h10q/l21_reducible_locus.py
nice -n 19 python3 math/h10q/l21_irred_generic.py
nice -n 19 python3 math/h10q/l21_irred_direct.py
```

The lead's replay measured respectively 0.3 seconds, 101 seconds, and 4 seconds. The commands regenerate `data/l21_reducible_locus.jsonl`, `data/l21_irred_generic.jsonl`, and `data/l21_irred_direct.jsonl`, together with their reports under `/tmp`; no refusal is used as evidence.

Run the L22 chain from the repository root, one command at a time, with the exact commands

```
nice -n 19 python3 math/h10q/l22_elimination.py
nice -n 19 python3 math/h10q/l22_reciprocal_cube.py
nice -n 19 python3 math/h10q/l22_infinity.py
nice -n 19 python3 math/h10q/l22_factor_tuple.py
nice -n 19 python3 math/h10q/l22_square_branch.py
nice -n 19 python3 math/h10q/l22_fiber_geometry.py
```

The respective output pairs are

Data artifact	Human-readable report
data/l22_elimination.jsonl	/tmp/l22_elimination.md
data/l22_reciprocal_cube.jsonl	/tmp/l22_reciprocal_cube.md
data/l22_infinity.jsonl	/tmp/l22_infinity.md
data/l22_factor_tuple.jsonl	/tmp/l22_factor_tuple.md
data/l22_square_branch.jsonl	/tmp/l22_square_branch.md
data/l22_fiber_geometry.jsonl	/tmp/l22_fiber_geometry.md

The strict licenses, in that order, are: PROVED uniform theorem with EVIDENCE scans; PROVED independent theorem with EVIDENCE scans; PROVED local structure with OPEN route endpoint; PROVED criterion/no-go with OPEN compatibility; PROVED free-parameter theorem but OPEN/excluded for the six-count; and PROVED reductions with an OPEN uniform lemma and EVIDENCE scans.

Run the L23 chain from the repository root, one process at a time:

```
nice -n 19 python3 math/h10q/l23_half_sieve.py
nice -n 19 python3 math/h10q/l23_fibration.py
nice -n 19 python3 math/h10q/l23_norm_section.py
nice -n 19 python3 math/h10q/l23_rational_section.py
nice -n 19 python3 math/h10q/l23_squareclass.py
nice -n 19 python3 math/h10q/l23_multivar.py
nice -n 19 python3 math/h10q/l23_absorption.py
```

The output pairs are

Data artifact	Human-readable report
data/l23_half_sieve.jsonl	/tmp/l23_half_sieve.md
data/l23_fibration.jsonl	/tmp/l23_fibration.md
data/l23_norm_section.jsonl	/tmp/l23_norm_section.md
data/l23_rational_section.jsonl	/tmp/l23_rational_section.md
data/l23_squareclass.jsonl	/tmp/l23_squareclass.md
data/l23_multivar.jsonl	/tmp/l23_multivar.md
data/l23_absorption.jsonl	/tmp/l23_absorption.md

Their strict boundary is not a chain from a sieve row to a full member: the small-prime lower bound is PROVED, the two-large-divisor sector is OPEN, the exact norm and Capell no-gos are PROVED, and finite searches are at most EVIDENCE.

Run the L24 chain under the same one-process policy:

```
nice -n 19 python3 math/h10q/l24_diagonal_geometry.py
nice -n 19 python3 math/h10q/l24_diagonal_arithmetic.py
nice -n 19 python3 math/h10q/l24_diagonal_local.py
nice -n 19 python3 math/h10q/l24_diagonal_search.py
```

The output pairs are

Data artifact	Human-readable report
data/l24_diagonal_geometry.jsonl	/tmp/l24_diagonal_geometry.md
data/l24_diagonal_arithmetic.jsonl	/tmp/l24_diagonal_arithmetic.md
data/l24_diagonal_local.jsonl	/tmp/l24_diagonal_local.md
data/l24_diagonal_search.jsonl	/tmp/l24_diagonal_search.md

The geometry, arithmetic, and local files carry PROVED exact statements. The search file replays those certificates, but its bounded zero-hit total remains EVIDENCE. In particular the unscaled diagonal five-count is PROVED FALSE by the dyadic theorem, not left OPEN by the scan.

Run the L25 scaled-coupling certificate:

```
nice -n 19 python3 math/h10q/l25_scaled_coupling.py
```

It writes `data/l25_scaled_coupling.jsonl` and `/tmp/l25_scaled_coupling.md`. The eliminants, parity classification, symbolic target formula, and named guarded real samples are PROVED; every complete small-prime residue scan and rational-point search is bounded EVIDENCE. The artifact does not weaken L24 and does not improve the conditional count.

Run the L26–L34 producers:

```
nice -n 19 python3 math/h10q/l26_reciprocal_tie.py
nice -n 19 python3 math/h10q/l27_triangular_shear.py
nice -n 19 python3 math/h10q/l28_trace_field.py
nice -n 19 python3 math/h10q/l29_reciprocal_frontier.py
nice -n 19 python3 math/h10q/l30_quartic_frontier.py
nice -n 19 python3 math/h10q/l31_frontier_push.py
nice -n 19 python3 math/h10q/l32_local_parameter_frontier.py
nice -n 19 python3 math/h10q/l33_unweighted_pair_main.py
nice -n 19 python3 math/h10q/l34_ap1_pair_threshold.py
```

They write the corresponding artifacts

`data/l26_reciprocal_tie.jsonl`
`data/l27_triangular_shear.jsonl`
`data/l28_trace_field.jsonl`
`data/l29_reciprocal_frontier.jsonl`
`data/l30_quartic_frontier.jsonl`
`data/l31_frontier_push.jsonl`
`data/l31_magma_genus2.json` (external audit)
`data/l32_local_parameter_frontier.jsonl`
`data/l33_unweighted_pair_main.jsonl`
`data/l34_ap1_pair_threshold.jsonl`.

Matching reports are written under `/tmp`. L31's $w = 13$ row is a theorem because it carries the full lift, primality, and Hilbert certificate; the remaining L29 member rows retain their finite EVIDENCE license. The external Magma [2] artifact records exact code, version, complete point sets, and intrinsic-help semantics; it has no standalone checked-in Magma certificate. L27 and L30's finite target cross-checks accompany uniform character proofs. L28's theorem is its exact Newton/resolvent/norm proof; L29's is its Hensel/congruence/resultant proof.

6. Review-driven correction as methodology

The L17 cycle treated review objections as attacks on the estimand, not as requests for cosmetic caveats. Five corrections illustrate the method.

Recurring-prime-only support. The first small-prime product used only primes recurring in the observed sample. Omitting nonrecurring bad primes mechanically inflated the predicted clean mass; it was not evidence of cross-prime correlation. The fix was to construct closure-authority root tables across every prime in the declared window and to target larger primes actually observed in the same cell, with simple roots assigned exact mass $1/(p+1)$ and singular roots recursively resolved or intervalized.

Wrong-sign conditioning. A proposed explanation attributed the clean-rate offset to conditioning on prime Q, but the sign was backwards. At the Q-root residue, $2\alpha b \equiv 0$, so the implementation marks the residue safe rather than bad. The corrected interpretation is a difference between Haar measure on $\mathbb{Z}_p$ and the finite k-window. Its measured size is approximately $+0.0195$, or 3.63 standard deviations in the artifact; it is characterized, not promoted to a dependence claim.

Cohort versus closure progression. The first matcher combined sieve members rebuilt from canonical closure classes with roots computed on an earlier wave cohort. In the documented ESC example at cell $(3,(-3,1))$, the cohort used $q_1 = 41$, $\varepsilon = +1$, whereas the closure used $q_1 = 59$, $\varepsilon = -1$.

The fix was to rebuild every root in the closure record's own (ε, f, q_1, N) coordinate. The repaired table reconciles 5,375/5,375 observed odd-negative occurrences.

Discarded resolved mass. An excluded singular root can have a rigorously resolved prefix even when its remaining Taylor tree is unfinished. An intermediate matcher discarded that known mass along with the unresolved remainder. The corrected artifact stores both pieces and forms the interval

$$\left[\prod_p (1 - k_p - r_p),\ \prod_p (1 - k_p)\right].$$

Across the 67 excluded pairs the maximum unresolved residual is 6.9×10^{-15}, so the finite-window interval is valid without pretending to control any larger prime.

The vacuous integer-support p-value. A clipped two-sided empirical value on discrete integer support reported 1.0, which carried little diagnostic information. The final report keeps the marginal-preserving null but presents its standardized statistic and both empirical tails. Together with the repaired support and masses, the final matched values are marginal ratio 0.9993, union residual $-2.9/8760$, $z = -0.09$, and tails 0.61/0.53. These are the final EVIDENCE numbers; the abandoned instruments license nothing.

7. Limitations

(1) Five older L17 files are persisted evidence without checked-in original producers: `l17_sieve.jsonl`, the cohort `l17_badroots.jsonl`, `l17_ratemodel_censored.jsonl`, `l17_cofactor.jsonl`, and `l17_cofactor_nonclosure.jsonl`. Downstream scripts consume reviewed copies, but the original session-side acquisition scripts were not recovered.

(2) The class factors and matched Haar comparison are statistics only for $p \le 10^4$. They provide no bound whatsoever for primes above 10^4, so they cannot by themselves prove a positive infinite product or a uniform density.

(3) The nonclosure cofactor distribution is measurement-limited. Only 11/400 sampled rows were exactly resolved and 389 were prover refusals; the predeclared power gate failed. The distributional conclusion is therefore INCONCLUSIVE, even though parity remained consistent on resolved rows.

(4) Some closure records contain auxiliary tied-status or ramification refusals. For example, the H109 closure records budget/refusal values in those fields while its proved zero-bad ladder verdict remains the accepted closure form. Such fields must not be silently rewritten as success, and their refusals are not supporting evidence.

(5) L22 removes the former vertical-fiber premise but does not prove simultaneous prime values. For every fixed $Z \in \mathbb{Q}^\times$, the canonical octic is irreducible and Hilbert selection produces an admissible a; classical Schinzel H is still required for the resulting linear–octic pair. The finite off-grid horizon proves neither Schinzel nor an unconditional all-cell member theorem.

(6) The L23 beta-sieve count is unconditional but stops at small-prime cleanliness. Degree-eight values may retain 16 factors above $X^{49/100}$, and reciprocity permits two or more bad factors. Neither the finite-window data nor the positive lower bound proves one fully emergent-free member. The first missing estimate is the two-large-bad-divisor sector beyond Bombieri–Vinogradov.

(7) The L24 unscaled diagonal route is not an unresolved possible improvement. It is PROVED FALSE on Φ: both unit orientations are empty over $\mathbb{Q}_2$. The separately OPEN absolute surface-classification question cannot restore a rational point to an empty local image, and the bounded search is not the proof.

(8) L25 shows that this is not a universal coupled-formula wall. The scaled type-I couplings are locally admissible on complementary s-parities, but global rational points, aligned-target

compatibility, and the $w = 3, 5$ analysis remain OPEN. No member theorem or improved count follows.

(9) L26 changes the algebraic coordinates, not the theorem: its reciprocal octic has a quartic trace character and automatic dyadic splitting. L28 proves the trace quartic and bad-sign cover uniformly irreducible. L29 now classifies the distinct reciprocal lift exactly and supplies a target-preserving compressed pullback, but no global reciprocal member theorem.

(10) L27 closes the displayed local coupling gaps on a new formula. One shear covers both dyadic parities and every standard aligned odd target, including $3, 5$. The linear-B, canonical-λ, and $y = 0$ sections are excluded, but no rational point on the full global octic is known.

8. Conclusion

L22 proves irreducibility on the fixed canonical branch for every rational $Z \neq 0$. Quantitative Hilbert irreducibility selects a good a in the existing character progression, so every L20 admissibility condition is PROVED.

L23 changes the unconditional boundary, not the theorem. The half-dimensional sieve gives $\gg X/(\log X)^{3/2}$ small-prime-clean progression members, but its exact parity audit leaves $0, 2, \ldots, 16$ possible large bad factors. The conic-bundle model has non-split rank 10 and Brauer quotient $\mathbb{Z}/2$, while the audited norm, Capell, fixed-twist, and factor-section escapes either fail on Φ or reduce to the named higher-genus and character obligations.

L24 closes the unscaled diagonal proposal negatively. Both exact degree-eight unit orientations are Φ-empty over $\mathbb{Q}_2$; the formal at-most-five rank count for that proposal is therefore vacuous. L25 proves that fixed scaling removes the universal dyadic wall, but leaves parity and aligned-target issues in those formulas. L26 gives a different reciprocal tie with a quartic trace character and automatic dyadic splitting. L28 proves its trace field and bad-sign extension uniformly nontrivial; L29 classifies its distinct lift and adds a target-preserving compressor plus a fixed-field slice. L27 gives a single shear covering both dyadic parities and every aligned odd target. L30 then replaces its degree-eight equation by a lower-degree constant-two even quartic with a selected fibre at every odd target and proves broad generic-section rigidity. L31 proves one exact fixed-field reciprocal member and one rational quartic point before the cube pullback. L32 gives automatic-Φ local bases at 2 and every odd target and closes two natural section menus. L33 proves the positive unweighted pair main. L34 identifies the unrestricted root-pair envelope that would imply AP1. Emergent map primes, that conditioned prime-weighted root moment, the uniform member theorem, and a cube-compatible controlled quartic point remain OPEN; no count improvement is claimed.

The established implication is unchanged: the L20 class feeds the L19 linear–octic pair; classical Schinzel H supplies simultaneous prime values; reciprocity gives an emergent-free member; and L6 gives a six-universal-quantifier definition of $\mathbb{Z}$ in $\mathbb{Q}$ with $\mathrm{efd}_{\mathbb{Q}}(\mathbb{Q} \setminus \mathbb{Z}) \leq 5$. Schinzel H remains the sole conjectural input. The result is conditional and does not solve H10($\mathbb{Q}$) unconditionally.

The exact next frontier is a uniform fixed-field reciprocal member theorem, a cube-compatible controlled point on L30's quartic, or AP1 via a pointwise fixed-family Buchstab/Hilbert-detector asymptotic. All 24 L22–L31 generators rebuild their named artifacts; SHA-256 manifests detect drift, and strict labels keep finite EVIDENCE and route-local OPEN statements out of the PROVED closure. No new unconditional quantifier record is claimed.

References

[1] R. Crandall and C. Pomerance, *Prime Numbers: A Computational Perspective*, 2nd ed., Springer, New York, 2005.

[2] W. Bosma, J. Cannon, and C. Playoust, *The Magma algebra system I: The user language*, Journal of Symbolic Computation **24** (1997), nos. 3–4, 235–265; computation replayed with Magma V2.29-9.

[3] G. Jaeschke, *On strong pseudoprimes to several bases*, Mathematics of Computation **61** (1993), no. 204, 915–926.

Table 2. Principal artifact schemas.

Artifact	Row types and key fields	What the row licenses
`l17_sieve.jsonl`	`meta`; `class-summary` with `cell`, `family`, `members`, forbidden residues, rates, alarms; `global-summary`	Per-class residue determination and the recorded member window; aggregates are measured summaries, not tail bounds.
`l17_badroots_closures.jsonl`	`meta`; `class` with progression parameters, root lists, exact or residual masses, exclusions; `summary` with reconciliation	Certified bad-signed $k \bmod p$ roots and their stated p-adic masses for the closure progression; excluded roots license only an interval.
`l17_matched.jsonl`	`meta`, `meta-methods`, `global`, `class`; keys include matched support, expectations, residuals, window factors, permutation null, and partial flags	EVIDENCE for fit on the identical member sample. It licenses neither cross-prime independence nor behavior above the recorded prime window.
`l17_classfactors.jsonl`	Untagged per-class rows: `cell`, progression parameters, `partial`, `factor_lower`, `factor_upper`, `factor_window`, leading bad primes	Exact bookkeeping for each finite window interval; closure-effort correlation is descriptive EVIDENCE only.
`l17_schinzel_audit.jsonl`	`meta`; `canonical` with the two polynomials, coefficients, irreducibility certificate, value gcds, local counts; `residual` with pointwise witness and `schinzel_applicable=false`	Finite irreducibility, fixed-divisor, pair-gcd, and local-condition checks. Residual rows are pointwise only; no simultaneous prime values follow.
`l17_horizon*.jsonl`	Usually `meta`, `class-attempt`, `aligned-class`, `member`, `kernel-replay`, an untagged closure row, and `summary`; the compact H107 file has only meta and closure	Only a verified replay plus the exact closure row licenses an off-grid closure. Attempt and refusal rows are provenance, not negative evidence.
`l13h_all_closures.json`	Top-level `n_cells` and `records`; each record has `family`, `cell`, `a`, `eps`, `f`, `q1`, `N`, `k_zero`, `Q`, `rung`, `tied`, `ramified_empty`, `source`	The canonical closure authority. Its thirteen-field record licenses the zero-bad ladder closure; auxiliary tied or ramified fields may themselves be refused without weakening that license.

Table 3. L19–L21 theorem and frontier artifacts.

Artifact	Row types and key fields	What the row licenses
`l19_tauzero.jsonl`	`meta`; algebraic-criterion and uniform-theorem rows; `uniform-witness`; bounded tuple audits; summary	The factor-by-factor $\tau = 0$ wall criterion and existence of a class-side break for every canonical 5-wall. Bounded aligned calls do not license members outside their recorded rows.
`l19_classexist.jsonl`	`meta`; per-cell generalized certificates; canonical collision rows; character and CRT counts; summary	The exact residue construction underlying uniform class existence and its finite corroborations. It licenses no prime value of the octic and no emergent-free member.
`l19_cell179.jsonl`	`meta`; `class-attempt`; `aligned-class`; `member`; exact closure; summaries and refusal counts	The replayed ladder-zero closure of $[179, [-1, 1]]$. Attempts and prover refusals remain provenance, not negative evidence.
`l20_admissible.jsonl`	`meta`; normalized per-class polynomials and content; irreducibility certificates; congruence checks; counterexample and summary	Uniform proofs of admissibility (b), (c), (d), and positive lead in (a); per-row irreducibility at the L20 stopping point.
`l21_reducible_locus.jsonl`	`l21a`; `l21b`; `palindrome`; `summary`, with factor degrees, square-class identities, endpoint equality, and controls	The explicit quartic-by-quartic reducible locus, its uniform exclusion from $\tau^\dagger$, and the near-palindromic identity.
`l21_irred_generic.jsonl`	`meta`; `generic-specialization`; `grid-fiber-certificate`; `first-irreducible-admissible-a`; `reducible-hunt`; summaries	Irreducibility over $\mathbb{Q}(a, Z)$, the 353 separate vertical-fiber certificates, quantitative Hilbert selection on those fibers, and exact finite hunt rows. Its L21-only scope left an uncertified z OPEN; the uniform L22 theorem supersedes that premise.
`l21_irred_direct.jsonl`	`meta`; 353 `constructed-grid-pair` rows; `summary` with reductions, Newton polygons, trace data, and auxiliary certificates	The three local no-go laws, reciprocal-iff-$a = 1$, and the trace-quartic and nonsquare-lift results. Its L21 blanket verdict remained OPEN; the independent L22 proofs close the fixed canonical branch.

Table 4. L22 data artifacts, checked-in producers, and strict licenses.

Data artifact	Producer	License
`l22_elimination.jsonl`	`l22_elimination.py`	PROVED: fixed-Z irreducibility on $(\tau^\dagger, \delta) = ((1 + 2a^2)/A, -4a^4/A)$ for every $Z \in \mathbb{Q}^\times$, including the $Z = 1$ certificate. EVIDENCE: the finite-field diagnostic tables.
`l22_reciprocal_cube.jsonl`	`l22_reciprocal_cube.py`	PROVED: on $(a, A, s, \tau^\dagger, \delta) = (1, 5, 0, 3/5, -4/5)$, trace irreducibility and the complete elliptic descent prove the octic irreducible for every $Z \in \mathbb{Q}^\times$. EVIDENCE: bounded searches and fiber scans.
`l22_infinity.jsonl`	`l22_infinity.py`	PROVED: exact infinity and zero-place clusters. OPEN for this route: local data alone still permit factor degrees 2, 4, and 6; it is not the closure.
`l22_factor_tuple.jsonl`	`l22_factor_tuple.py`	PROVED: weakened factor-tuple criterion and reciprocity no-go. OPEN for this route: an all-plus compatibility theorem; parity alone controls only the product of outside signs.
`l22_square_branch.jsonl`	`l22_square_branch.py`	PROVED: free-λ algebraic and class theorems. OPEN/excluded for the six-count: L11c makes an infinite moving branch parameter an additional witness.
`l22_fiber_geometry.jsonl`	`l22_fiber_geometry.py`	PROVED: Noether reduction, fixed-fiber descent, and generic geometric statements. OPEN for this route: the uniform polynomial nonvanishing lemma; residue scans are EVIDENCE.

Table 5. L23 data artifacts, checked-in producers, and strict licenses.

Data artifact	Producer	License
`l23_half_sieve.jsonl`	`l23_half_sieve.py`	PROVED: dimension $1/2$, Bombieri–Vinogradov level, and $\gg X/(\log X)^{3/2}$ members clean below $X^{49/100}$. OPEN: removal of two large bad divisors; no full-member theorem.
`l23_fibration.jsonl`	`l23_fibration.py`	PROVED: closed degrees $1, 8, 1$, non-split rank 10, $\mathrm{Br} / \mathrm{Br}\,\mathbb{Q} = \mathbb{Z}/2$, and the refined Capell obstruction. OPEN: the named fixed even-$v_w(z)$, $w \mid s$ stratum and unconditional points.
`l23_norm_section.jsonl`	`l23_norm_section.py`	PROVED: exact-match and polynomial-in-b norm no-gos, with the surviving curves isolated. OPEN: rational points and sections outside the classified ansatzes.
`l23_rational_section.jsonl`	`l23_rational_section.py`	PROVED: no generic b-line section and exact genus-4 factor residuals. OPEN: isolated per-cell points and nonconstant twists.
`l23_squareclass.jsonl`	`l23_squareclass.py`	PROVED: $P = \pm 2b$ times a square is Φ-empty, $P = A\,\mathrm{Norm}$ has dyadic symbol -1, and the reciprocal Capell locus is empty. Bounded searches are EVIDENCE.
`l23_multivar.jsonl`	`l23_multivar.py`	PROVED: exact degree formulas for the tested moving families and a squarefree degree-$18 = 2 + 16$ certificate. No minimum 21 and no unconditional closure are licensed.
`l23_absorption.jsonl`	`l23_absorption.py`	PROVED: zero-cost identifications, finite-flat rank bound, and dyadic emptiness of the diagonal image. OPEN: record closure; the formal at-most-five count is vacuous.

Table 6. L24 data artifacts, checked-in producers, and strict licenses.

Data artifact	Producer	License
`l24_diagonal_geometry.jsonl`	`l24_diagonal_geometry.py`	PROVED: both finite-flat degree-8 orientations are Φ-empty; arithmetic monodromy is $V_4 \wr C_2$ of order 32, the audited b-slices have genus 35, and there is no base-rational section. Absolute surface rationality remains separately OPEN.
`l24_diagonal_arithmetic.jsonl`	`l24_diagonal_arithmetic.py`	PROVED: the complete orientation-I valuation proof, orientation-II modulo-16 proof, and genus-53 z-slices. The union is empty and the diagonal route is CLOSED.
`l24_diagonal_local.jsonl`	`l24_diagonal_local.py`	PROVED: orientation II is locally soluble at every odd target in the standard $k = 1$ stratum, while both orientations are globally superseded by the dyadic obstruction.
`l24_diagonal_search.jsonl`	`l24_diagonal_search.py`	PROVED: exact replay of the imported dyadic certificate on each row. EVIDENCE: 4,026,282 bounded attempts and zero rational hits; the no-hit total is not the uniform proof.

Table 7. L25 data artifact, checked-in producer, and strict license.

Data artifact	Producer	License
`l25_scaled_coupling.jsonl`	`l25_scaled_coupling.py`	PROVED: exact scaled eliminants; $(2X, \rho)$ is 2-adically admissible exactly on even s, and $(2X, \rho/2)$ exactly on odd s; a symbolic unit-b target formula with (ρ, X)-Jacobian -2400 for all $w \nmid 2400$; guarded parity-wise real samples. OPEN: global rational points, aligned-target compatibility, and the target primes $3, 5$. Bounded scans and searches are EVIDENCE.

Table 8. L26–L34 data artifacts, producers, and strict licenses.

Data artifact	Producer	License
`l26_reciprocal_tie.jsonl`	`l26_reciprocal_tie.py`	PROVED: reciprocal quartic-trace identity, dyadic square, guarded W1 target, and trace-character descent. L28 supersedes the former conditional trace-field license. L31 promotes the $w = 13$ fixed-field row; no uniform member theorem follows.
`l27_triangular_shear.jsonl`	`l27_triangular_shear.py`	PROVED: exact shear octic, one-piece dyadic Hensel theorem, all-odd-prime aligned-target theorem, one guarded real sample, and three scoped section no-gos. OPEN: full global rational point, member theorem, and five-count.
`l28_trace_field.jsonl`	`l28_trace_field.py`	PROVED: uniform trace-quartic irreducibility over $\mathbb{Q}_2$, the norm squareclass $N(2(\theta+2)) \in A\mathbb{Q}_2^{\times 2}$, and irreducibility of the bad-sign Capell cover. L29 supersedes the distinct-lift question. L31 proves one member; the uniform parity step and every count improvement remain OPEN.
`l29_reciprocal_frontier.jsonl`	`l29_reciprocal_frontier.py`	PROVED: complete dyadic reciprocal-lift classification, ordinary resultant certificate at $v_2(Z) = -2$, target-preserving compressed pullback, and fixed-field reduction. L31 promotes the $w = 13$ row with a complete certificate. The other finite fibres and fixed-row parity scan remain EVIDENCE. OPEN: uniform member theorem and every count improvement.
`l30_quartic_frontier.jsonl`	`l30_quartic_frontier.py`	PROVED: exact constant-two even quartic, uniform dyadic Hensel theorem, fibre-regular rows for every odd $w \geq 5$, an exact strong-Hensel $w = 3$ fibre with an empty fixed-a control, one real stratum, generic square-branch/shear rigidity, and square-pullback trace-base/lift separation. The 13,224-row zero-hit probe is EVIDENCE. OPEN: cube-compatible controlled quartic root, member theorem, and every count improvement.
`l31_frontier_push.jsonl`; `l31_magma_genus2.json`	`l31_frontier_push.py`; external Magma V2.29-9 [2] replay	PROVED: one exact fixed-field reciprocal member at $w = 13$; the mandatory lift, recursive Pocklington chain, and complete Hilbert table; the unique constant-$c = -64/25$ bridge-square section and one exact rational H_2-point before the cube pullback; two genus-2 cube reductions; the positive-main dispersion correction; and AP1 equivalent to intermediate H inside the selected L19–L22 aligned-class/Q-prime protocol. The external `RationalPointsGenus2` completeness flags show both cube curves have only $z = 0$, closing that unique section. OPEN: other cube-compatible controlled points, uniform member theorem, AP1, and every count improvement.
`l32_local_parameter_frontier.jsonl`	`l32_local_parameter_frontier.py`	PROVED: automatic-Φ rational maps; a quartic character-sum theorem giving a smooth L30 base at every $w \geq 5$; a three-map b-menu with $v_w(b) = 1$; the non-uniformity of every finite fixed proportional-trace menu; and the restriction $b = 1 \pm 2a$ to $w \equiv 5, 19 \bmod 24$. OPEN: control of emergent numerator/denominator primes, a global H_2-root, AP1, and every count improvement.
`l33_unweighted_pair_main.jsonl`	`l33_unweighted_pair_main.py`	PROVED: for every $0 < \alpha < \beta < 1/2$, the unweighted bad-root pair sum is asymptotic to $\frac{1}{8}\log^2(\beta/\alpha)X$, by L23 Chebotarev and exact CRT counting. OPEN: transfer to $\Lambda(Q(t))$, exact odd valuations modulo p^2q^2, small-prime cleanliness, AP1, and every count improvement.
`l34_ap1_pair_threshold.jsonl`	`l34_ap1_pair_threshold.py`	PROVED: the exact root-oversieve inequality $\mathcal{N}_{\text{good}} \geq \#\mathcal{A}_z - \mathcal{P}_2^{\text{root}}$, the threshold $\vartheta > 8e^{-2\sqrt{2}}$, and the unrestricted $\vartheta = 0.49$ pair envelope $0.9749604961\ldots < 1$. OPEN: the conditioned prime-weighted root-moment estimate, AP1, and every count improvement. Exact odd valuations are not needed for this upper-bound route.